A. Robin Helps

time limit per test 1 second
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 800 | Tags: greedy, implementation | https://codeforces.com/problemset/problem/2014/A

There is a little bit of the outlaw in everyone, and a little bit of the hero too.

The heroic outlaw Robin Hood is famous for taking from the rich and giving to the poor.

Robin encounters n people starting from the 1-st and ending with the n-th. The i-th person has a_i gold. If $a_i \geq k$, Robin will take all a_i gold, and if $a_i=0$, Robin will give 1 gold if he has any. Robin starts with 0 gold.

Find out how many people Robin gives gold to.

Input

The first line of the input contains a single integer t ($1 \leq t \leq 10^4$) — the number of test cases.

The first line of each test case contains two integers n, k ($1 \leq n \leq 50$, $1 \leq k \leq 100$) — the number of people and the threshold at which Robin Hood takes the gold.

The second line of each test case contains n integers $a_1, a_2, \ldots, a_n$ ($0 \leq a_i \leq 100$) — the gold of each person.

Output

For each test case, output a single integer, the number of people that will get gold from Robin Hood.

Example

Input

```
4
2 2
2 0
3 2
3 0 0
6 2
0 3 0 0 0 0
2 5
5 4
```

Output

```
1
2
3
0
```

Note

In the first test case, Robin takes 2 gold from the first person and gives a gold to the second person.

In the second test case, Robin takes 3 gold and gives 1 gold to each of the next 2 people.

In the third test case, Robin takes 3 gold and so only gives gold to 3 other people.